CHAITHRA LOKASARA MAHADEVASWAMY
柴麗俐 (LILLY)

AI/ML Engineer | Google Cloud Certified Generative AI Leader | Medical AI R&D | Production-Grade MLOps | End to End ML systems | AI Product Ownership (Dentistry AI) | Software Team Coordinator (iDental) - Open for Relocation | Indian | Major Engineer of SmartsurgeryTek driving IDental Product | Indian- Open for Relocation



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PROFESSIONAL SUMMARY

Ownership-driven Machine Learning Engineer with 1+ year of industry experience from data strategy to model development, fine-tuning, and production ML systems deployment and monitoring, 3+ years of academic research NLP, GenAI, LLMs, and 1+ year of Healthcare R&D experience. Specialized in Computer Vision, Medical AI, Multimodal/Mutlilingual AI, RAG, and AI Agents, with experience delivering end-to-end AI solutions for production products. Skilled in building scalable data and inference pipelines, deploying ML systems, and collaborating across engineering, product, data, and clinical teams. Experienced in mentoring junior engineers and interns while taking ownership of AI product development and representing technical solutions across stakeholders. Coordinate the Software Foundation team across development workflows, CI/CD deployments, and operational issues to support reliable AI product delivery. Focused on building production-grade AI systems that contribute to product growth, business impact/ROI, and organizational success.

EDUCATION AND ACADEMIC HONORS

Chang Gung University
(長庚大學)
Taoyuan, 臺灣 R.O.C Taiwan
2023/09-2025/08

Master of Science in Artificial Intelligence (MS in AI)

Honors: Certificate of Merit (Academic Excellence), Outstanding Graduate Student Award (MOE/MOFA Taiwan), Taiwan Ministry of Education Scholarship Recipient for Masters degree.
Specialized in NLP and Generative AI with experience in data annotation, data mining, web scraping, and model fine-tuning. Research team lead at WIDElab with contributions to international conferences and journal publications.

Presidency University
Bangalore, India
2019/08-2023/07

Bachelor of Technology (B.Tech in Computer science and Engineering)

Honors: Awarded Abdul Hameed Memorial Gold Medal (Rank 1 / University Topper in CS)
Strong computer science foundation covering Operating Systems, System Engineering(C, C++, Java), Object-Oriented Programming (OOP) and OOAD, Data structures &v Algorithms , Time and space complexity, Probability & statistics, Software Project Management along with core knowledge of DBMS, Big Data systems, and IoT.

WORK EXPERIENCE

Smart Surgery Tek
慧術科技股份有限公司
Taipei city , R.O.C Taiwan
2025/10 - Present

Mid level Machine Learning Engineer

Key contributions

- Took full ownership of the iDental AI product, independently designing and delivering all AI-driven features / Software driven decisions across the dentistry AI pipeline, from model development to production deployment and monitoring
- Optimized the end-to-end inference pipeline, reducing average response latency from 11 seconds to 3 seconds - 73% speed improvement.
- Coordinate software engineering activities to meet delivery milestones and project requirements. Collaborate closely with Product Managers, own and manage the Jira board, assign and track tasks across the team, and oversee outsourced engineers and junior developers to ensure timely and efficient delivery. Implement Active learning Process for entire Data - ML AI Automated workflow
- Built and optimized medical imaging AI models (segmentation, detection, predictive AI) achieving 95%+ performance across key evaluation metrics for dentistry AI systems for TFDA/FDA - HIPAA GDPR compliant.

Unicorn Tutor AI
Taipei city , R.O.C Taiwan
2025/06 - 2025/08

AI Tools Research Intern

Key contributions

- Assisted with product testing, functional evaluation, and performance assessment of AI-powered applications.
- Optimized prompts to improve LLM response quality, consistency, and task accuracy.
- Built and executed a comprehensive evaluation framework for LLM and Speech AI systems (ASR/TTS), benchmarking models such as Whisper (large-v3-turbo), Azure Speech, and OpenAI APIs across accuracy, latency, robustness, and multilingual/noisy conditions.
- Designed and integrated end-to-end AI testing pipelines with prompt engineering, phoneme-level speech evaluation (PER scoring), and API-based automation to assess model performance and support data-driven model selection for product use cases.

Chang Gung Univeristy
長庚大學
Taoyuan, R.O.C Taiwan
2023/11 - 2025/08

Research Assistant

Key contributions

- Contributing to Journals and conference research projects. Owning Hugging face team for WIDElab.
- Developed NLP pipelines for sequence-to-sequence modeling, sentiment analysis, and aspect-based sentiment analysis (ABSA)
- Built and optimized RAG-based applications using vector databases and LLM APIs

National Taiwan Normal Univeristy
國立臺灣師範大學
Taipei city, R.O.C Taiwan
2022/11 - 2023/4

Robotics Research Intern

Key contributions

- Winning IROS IEEE HAC Humanoid challenge in 3rd place as a Team Lead.
- Worked on an Intelligent Humanoid Robot (ROBOTIS-OP3) at the Education Robotics Lab focusing on humanoid robotics research and development, Applied ROS, computer vision, sensor fusion, machine learning, and deep learning for real-world robotic perception and control tasks

CONTACT & LINKS 📍 TAIWAN 📧

🔗 **LinkedIn Profile :** https://www.linkedin.com/in/chaithra-lokasara-mahadevaswamy-5bb076214/
💻 **GitHub:** https://github.com/Chaithra-lm
Hugging Face : https://huggingface.co/Chaithra26
Credly : https://www.credly.com/users/chaithralm/badges/credly

AI TOOLS THAT I USE EVERYDAY

AI-Assisted Development & Productivity

- Claude Code/Codex
- GPT/Gemini Pro
- Antigravity/Amazon Q
- Jira
- GitHub Copilot
- VS code

Developer tools

- Git/Github Actions
- Postman/Insomnia
- Jupyter/Colab Notebooks

AI Research tools and Data Exploration

- Gemini Notebook/Code Assist/Workspace
- Hugging Face/ollama
- Looker/Julius AI/Notion

Cloud Native Tools/ Infrastructure Knowledge

- AWS Sagemaker AI/Auto ML
- ML Flow
- Vertex AI / AI studio/Agent studio
- Amazon S3/ECS/ECR/EKS/EC2/Cloudwatch/EventBridge/Security agent
- GCS/GCP Compute Engine/BigQuery/Pretrained API/Google Analytics 4
- Artifact Registry, Cloud Run, API Gateway

TECHNICAL SKILLS

- **Machine Learning & Deep Learning :** Supervised Learning, Reinforcement Learning, Sequence-to-Sequence Modeling, Feature Engineering, DNNs, CNNs, Predictive AI solutions, Transfer Learning , Model Training & Fine-tuning, Hyperparameter Optimization, Data Augmentation, Classification, Regression
- **Computer Vision:** Image Classification, Image Segmentation, Semantic Segmentation, Instance Segmentation, Object Detection, Keypoint Detection, Pose Estimation, Medical Imaging, Dental & Surgical AI, Panoramic X-ray Analysis, Peri Apical X-ray, Radiology, Robot Perception, Auto-Annotation , building Computer vision utilities using Trained Models, Image Preprocessing & Postprocessing, DICOM Standards Processing and training
- **Natural Language Processing & Generative AI:** Multimodal & Multilingual Large Language Models (LLMs), PEFT (Parameter-Efficient Fine-Tuning) LoRA/QLoRA, Transformer Models, Prompt Engineering & Optimization, Zero-Shot & Few-Shot Learning, Chain-of-Thought (CoT) Reasoning, Prompt Chaining, ReAct Prompting, Text Embeddings, Semantic Search, Text Classification & Analysis, Aspect-Based /Target based Sentiment Analysis (ABSA), Generative Adversarial Networks (GANs).
- **AI Agent Tooling** : AI Agents, Agent Tooling, Harness Engineering, Agent Development Kit (ADK), MCPs, Multimodal Retrieval-Augmented Generation (RAG), Grounding Techniques, Vector Search 2.0, Vector Databases (Qdrant, Pinecone), n8n , AI automated workflow, LangChain , API Extension, Plugins, Cloud Functions, Google pretrained APIs
- **Mutimodal AI:** VLMs **,** Diffusion Models, Stable Diffusion, Image Generation Models, Image-to-Text Summarization, Image Captioning , Veo , Imagen
- **NLP & Transformer Models:** BERT, RoBERTa, Twitter-RoBERTa, XLM-RoBERTa, ViT (Vision Transformers Swin Transformer)
- **LLMs:** Qwen2.5, Mistral 7B, Llama 3.1 70B, OpenAI GPT Models, Nemotron 3 models/Nemoclaw
- **Speech & Voice AI:** Automatic Speech Recognition (ASR), Text-to-Speech (TTS), Real-time TTS, Conversational Systems, Speech Recognition & Evaluation, Whisper (OpenAI), MeloTTS, Coqui TTS, ElevenLabs, Microsoft Azure Speech Services, SpeechSuper, Phoneme-level Pronunciation Analysis, Phoneme Error Rate (PER) Evaluation
- **Data Engineering : Data Versioning DVC, Data Warehouse , Data Lake , Data lake house(Data bricks) ,** Data Storage & Management (Amazon S3, Google Cloud Storage), ETL (Extract, Transform, Load), Data Pipelines, Exploratory Data Analysis (EDA), Data Mining, Data Preprocessing, Data Cleaning, Data Validation, Dataset Creation & Curation, Data Annotation (Text & Image), Synthetic Data Generation, Data Quality Assurance, PII Data Handling, Anonymization & Data Privacy
- **Databases & Data Technologies: AWS S3, GCS,** MySQL, SQL, Apache Hive, HBase, NoSQL/MongoDB, Amazon DynamoDB, BigQuery (Bigdata), Tableau
- **Frameworks & Core Libraries:** PyTorch, TensorFlow, Scikit-learn, Hugging Face (Models & Datasets), Ultralytics YOLO, OpenCV, CVAT, MONAI (Active learning), timm, Grad-CAM, Optuna, CVAT SDK, Pydantic, Boto3, Pandas, NumPy, SciPy, Matplotlib, Seaborn, OpenAI API, Gradio, Ollama
- **Deployment & Inference:** Stateful/Stateless Applications/Containers, Serverless Asynchronous Inference, **Model Versioning**, Model Deployment,Model Lifecycle Management, Environment Management (INT/STG/PRD), NVIDIA NIM, API hosted NIM(build.nvidia.com), Model Serving, Production Inference Pipelines, Model Benchmarking, Accuracy & Latency Benchmarking, Model Optimization (FP16/FP32), Concurrency Testing, Load & Stress Testing, Robustness Testing
- **Compute Infrastructure & Hardware:** CUDA, NVIDIA GPUs (L4, RTX 3090, A30, A100, A6000), AMD GPUs, ROCm Ecosystem
- **Inference/Cost/Latency Optimization & Acceleration:** GPU Optimization, TensorRT, ONNX Runtime, DirectML, AWS Neuron SDK, Inferentia1/Inf1, Inferentia2/Inf2, Trn1, Model Compilation & Optimization , resource scaling, cost optimization **Containerization & Orchestration:** Docker, Kubernetes (K8s), Helm, Terraform, Ansible
- **Robotics Middleware:** Robot Operating System(ROS), Motion Planning, Sensor Fusion (Camera IMU processing), Human-Robot Interaction (HRI)
- **Operating Systems & Systems:** OS, Ubuntu, Windows, macOS, Linux kernels, Shell Scripting/CLi Bash commands.
- **Git & GitHub:** Git Commands & Branching, GitFlow, GitHub Flow, Pull Requests & Code Reviews, Advanced Rebase, Cherry-pick, Reflog, GitHub Actions, Reusable Workflows, Git Secrets, Semantic Versioning (SemVer), Periodic Release Management, Git LFS, DVC, Production Deployment & Rollback.
- **Software Development Methodologies : Agile Project Management · Scrum · Kanban · Sprint Planning · Product Backlog Management · Requirements Engineering · User Story Mapping · Stakeholder Management · Cross-Functional Collaboration · Iterative Delivery, Retrospective Facilitation · Workflow Optimization · Risk and Dependency Management,** Product Management, Jira Project Management, Code Reviews for Junior engineers, Jira Board Management, Ticket assignment Ownership. Troubelshooting
- **Software Engineering/Testing:** Python (Primary), C++, Java, C, Go, R, Software Debugging, Pytest, Unit Testing, Integration Testing, Automated Testing, Software Requirements Specification (SRS), Business Logic, CI/CD Pipelines , Oversaw feature Integration development to staging, staging to Production.
- **Web Development/System/API Integration:** HTML5, CSS3, JavaScript/React.js, FastAPI, REST APIs, Schemas, Microservices, API Integration, AWS API Gateway, Postman, Insomnia,OpenAPI/Swagger, JSON, HTTP/HTTPS, Backend systems , Client Dashboard/Reports, HIS/workstation Integration, Multi Tenant SaaS/User roles,
- **Technical Documentation and Research :** Academic and Industrial Research Publications, AI/Software TFDA, US FDA Technical Documentation, Research Communication/Plans, Research paper selections for the problems solving, Research Team Leadership
- **AI Security/Cybersecurity:** Secure AI Systems, Cisco Cybersecurity fundamentals, SAIF, Threat Modeling, AI Penetration Testing, AI Risk Assessment, AWS Security agent
- **Medical AI Regulatory : HIPAA • GDPR** Product Documentation, Medical AI Validation, Clinical Risk Analysis, Regulatory Compliance, Traceability, Data governance (**IRB**)
- **Observability, Monitoring & Logging :** LGTM Stack (Loki, Grafana/Prometheus, Tempo, Tensorboard), AWS CloudWatch/Cloudtrail logs, Bias/Drift Monitoring, Metrics Monitoring, Performance/Efficiency Monitoring, Distributed Tracing, Production Log Analysis
- **Product development and Management :** Functional Testing, Test Case Design and Execution, Performance Testing, Product Requirements, Requirements Analysis, Product Feature Development, Feature Brainstorming, Product Testing, User Acceptance Testing (UAT).

PROFESSIONAL SKILLS

- Team Player/Team Lead/Coordinator
- Product Strategy/Delivery
- Accountable/Responsible
- Attention To Detail
- Sprint Planning
- Roadmap Planning
- Technical Documentation & Maintainability.
- Stakeholder/Team Communication
- Cross Functional Collaboration

RESEARCH PUBLICATIONS

1. C. L. Mahadevaswamy, J. Baltes, and H. -T. Chang, "**Humanoid Pose Estimation through Synergistic Integration of Computer Vision and Deep Learning Techniques," 2024 IEEE 9th International Conference on Advanced Robotics and Mechatronics (ICARM), Tokyo, Japan,** 2024, pp. 771-776, **https://doi.org/10.1109/ICARM62033.2024.10715957.**
2. C. L. Mahadevaswamy, K. Nguyen, M. Singh, and H. -T. Chang, **"Optimizing Retrieval in RAG Systems with Reinforcement Learning: A Trade-off Between Quality and Cost," 2025 Springer 9th International Conference on Natural Language Processing and Information Retrieval (NLPIR), Springer Nature Switerzland, Lecture Notes in Networks and Systems (LNNS), 2025, https://doi.org/10.1007/978-3-032-20897-2_49 (Best Paper Award 2025 December)**
3. Contributer Tech suppport and AI Expert guidance for **Dentistry AI**, **Springer Nature : Artificial intelligence-assisted diagnosis of dental caries and periodontitis on periapical radiographs: a clinical validation study. BMC Medical Informatics Q1 Journal (2026). https://doi.org/10.1186/s12911-026-03831-4**

VALIDATION AND CREDIBILITY

Key projects contribution

1. **Medical Image PII De-identification & Anonymization Utility (Privacy Gatekeeper)**
Machine Learning Engineer | Medical AI Security & Privacy | 2025-2026
Technologies: OCR, NER, Regex, Medical Imaging Compliance (PDPA/HIPAA)

- Architected a 3-stage automated pre-validation pipeline (OCR, Pattern Matching, NER) to detect Personally Identifiable Information (PII) in medical images prior to AI inference.
- Developed a medical image de-identification utility using CVAT to identify and anonymize Personally Identifiable Information (PII) embedded in clinical images prior to dataset creation and AI model training.
- Integrated OCR systems for high-precision text extraction and layout parsing across X-rays clinical images.
- Established a human-in-the-loop verification workflow through CVAT to review de-identified
- images and ensure that residual patient-identifying information was removed before downstream AI processing.
- Designed custom Regex rules for immediate pattern matching against Taiwanese National IDs, dates of birth, and medical record numbers.

2. **Peri-Apical X-ray Dental Diagnosis System (Periodontology, Caries)**
Machine Learning Engineer | Dental AI Project | 2025
Technologies: Radiology , ViTs, Object Detection, ML-Regression, Computer Vision, Medical Imaging

- Developed an AI system for automated periodontal measurements from peri-apical dental X-rays using ViTs
- Achieving FDA-approval performance metrics with over 90% accuracy.
- Collaborated with dental specialists to validate model predictions and clinical workflows
- **Contributer Tech suppport and AI Expert guidance for dentistry AI, Springer Nature :**
- **Artificial intelligence-assisted diagnosis of dental caries and periodontitis on periapical radiographs:**
- **a clinical validation study. BMC Medical Informatics Q1 Journal (2026)**.
- **https://doi.org/10.1186/s12911-026-03831-4**

3. **Panoramic Dental Dicom X-ray Diagnosis System (Artificial Material Segmentation, Periodontology , Caries , Periapical Lesion)**
Machine Learning Engineer | Dental AI Project | 2026-Present
Technologies: Dicom , Monai, Radiology, Object Detection, Object segmentation, Object Classification

- Developed an end-to-end diagnostic system for panoramic dental X-rays.
- Implemented object detection and image processing techniques for dental pathology analysis.
- Achieved TFDA-approved performance metrics exceeding 95%.
- Deployed production-ready inference pipelines for clinical usage.

4. **iOR – Intelligent Operation Room Documentation & Retrieval System**
Machine Learning Engineer | Healthcare AI Project | 2026–Present
Technologies: Python, FastAPI, LLMs, Multimodal RAG, LangChain, Qdrant, Gemini Embedding 2 / TwelveLabs Whisper, FFmpeg, AWS S3, PostgreSQL.

- Built an AI-powered system for operation room documentation, surgical video understanding, and automated summarization.
- Developed video and text ingestion pipelines with transcription, scene segmentation, metadata extraction, and semantic indexing.
- Implemented multimodal embeddings and vector-based retrieval using Qdrant for cross-modal clinical information search.
- Integrated RAG pipelines and LLMs to generate context-aware surgical summaries
r. retrieve relevant clinical information.

5. **Human-Governed Agentic Active Learning for Clinical AI**
Machine Learning Engineer | Healthcare AI / MLOps | 2026–Present
Technologies: Python, ZenML, CVAT, LangChain, LLM Agents, Active Learning, AWS SageMaker, Multimodal RAG

- Developed an end-to-end active learning workflow using ZenML for automated data preparation, model training, evaluation, and registry management.
- Integrated CVAT for human-in-the-loop annotation, reviewer approval, and clinical dataset quality control.
- Implemented agent-driven sample selection based on model confidence, prediction errors, data drift, and clinical feedback.
- Automated the continuous learning loop connecting production predictions, human review, dataset updates, and model retraining.

6. **Conversational AI Assistant (Voice AI agent) using Open AI API**
Independent Project | Python, LiveKit, OpenAI API, AsyncIO, Pytest | 2025

- Engineered a real-time multilingual voice and text AI agent with tool-calling capabilities.
- Implemented weather lookup, prompt rewriting, and semantic search using embeddings.
- Designed safe interaction flows with grounding, refusal handling, and function orchestration.
- Integrated OpenAI models:
  - GPT-4o-mini (LLM)
  - GPT-4o-transcribe / Whisper-1 (STT)
  - GPT-4o-mini-tts (TTS)
  - text-embedding-3-large (Embeddings)
- Implemented multilingual turn detection and voice activity detection (VAD).
- Achieved 100% pass rate on 7/7 end-to-end automated tests.

GitHub: AI Assistant LLM Agent Development Using LiveKit OpenAI

7. **Voice-Based Multilingual RAG System**
Independent Project | Voice AI, RAG, Vector Databases | 2025–Present (Personal project)

- Developed a real-time multilingual RAG pipeline using LiveKit for audio streaming.
- Enabled English speech input, Chinese document retrieval, and English speech output.
- Integrated Whisper, Sentence-BERT, Vector DB, GPT-4o, and MeloTTS.
- Developed a second version using Deepgram, OpenAI, and Cartesia voice agents.
- Delivered seamless speech-driven cross-lingual question answering.

Language Proficiency

- **English** C2 Proficiency (EF SET/TOEIC)
- **Chinese** (Mandarin) A2 Proficiency
- **Hindi** (Indian Native language)
- **Kannada** (Indian Native language)

Digital Training

- **Completed Introduction to Agile Management corporate training by Smartsurgery and HFU** covering Agile principles, Scrum, Kanban, sprint planning, requirements engineering, cross-functional collaboration, stakeholder management, and CI/CD.
- **Google and Google cloud technical session on AI agent builder and Healthteck Foundation models**
- **AWS SageMaker MLOps:** From Idea to Production – Hands-on training on building, deploying, and managing end-to-end ML pipelines using AWS SageMaker.
- **AWS Agent Security Training** – Learned to integrate agentic AI security into the Software Development Lifecycle (SDLC) using AWS security tools and best practices.
- **AWS Summit AI-DLC Training** – Training on the latest AWS AI/ML services, Generative AI, Deep Learning Containers (DLC), and cloud AI deployment.
- **Surgical Data Science (SDS) Summer School 2026** – Advanced program covering AI for surgical imaging, computer vision, medical robotics, and clinical AI research.
- **Global Summit on Transforming Healthcare with Digital Intelligence 2025**

Digital Certification and Badges

- **Google Cloud Certified Generative AI Leader Certification**
- **From Data to Insights with Google Cloud Specialization by Coursera**
- **Crash Course on Python by Google**
- **Databricks Acccredited - Generative AI Fundamentals**
- **Claude Code 101** from **Anthrophic**
- **Cisco Cybersecurity** essentials
- **Building LLM Applications With Prompt Engineering** from **Nvidia**
- **Data science Foundations** from **IBM**
- **ML - Linear Regression** by **Analyttica Datalab**
- **AI foundation** by **Nasscom Foundation** India
- **Certified Data Analyst** by **Whitehat Jr**
- Please follow my Linkedin Profile : To see check on all my **skills and acheivements certificates**

Awards & Honors

- **Best Paper Award – Springer** NLPIR 2025, Kyushu University (2025), Japan
- **Outstanding Graduate Student Award –** Ministry of Education & Ministry of Foreign Affairs Taiwan (2025) - MS degree
- **Certificate of Merit (Academic Performance) –** Chang Gung University (2025)
- **Abdul Hameed Memorial Gold Medal – B.Tech** Presidency University (2023) Topper in CS
- **Taiwan MOE Scholarship –** Ministry of Education Taiwan (2023)
- **3rd Place – Humanoid Application Challenge (HAC) –** IEEE IROS FIRA RoboWorld Cup and Summit (2023) , Taipei, Taiwan

Leadership and Service

- **Guiding Junior Interns and engineers at smartsurgery :** Technical Mentorship, Guiding Junior Engineers & Interns, **Code Reviews,** Technical Guidance, Knowledge Sharing, Onboarding & Task Support
- **Research Team Lead** at WIDE lab , CGU Taiwan
- **Class Representative –** B.Tech Computer Science & Engineering, Presidency University, india
- **Class Representative –** M.S. in Artificial Intelligence, Chang Gung University, india
- **Student Ambassador –** Taiwan AI Graduate Program, Chang Gung University
- **Student Ambassador –** Taiwan Higher Education Outreach, Presidency University
- **International Student Mentor/Counsellor –** Guided prospective and junior students on Taiwan scholarship applications, university admissions, and visa processes.
- **Task coordinator** : For Out-source Engineers